CSE 402 — Simulation & Random Numbers

Theory Revision Sheet

Topics (from **online3/**): Discrete-Event Simulation (single-server queue, *main focus*) · Linear Congruential Generators · Middle-Square & Weyl repair · RNG tests (χ^2, K–S, autocorrelation, runs) · Monte Carlo (integration, π, Buffon, reliability) · Inverse-transform variates. *Formulas in mathematical form; each method explains why it exists and how it compares to the one before it. Solved simulation problems at the end.*

The whole course in one line: GENERATE uniforms (LCG / middle-square) → TEST them (uniform? χ^2/K–S; independent? autocorrelation/runs) → USE them (inverse-transform to any distribution; Monte Carlo for integrals/π/reliability; discrete-event simulation for queues).

1 Discrete-Event Simulation — Single-Server Queue (main focus)

Idea. A *discrete-event* simulation advances the clock only to the instants when something happens (an *event*); between events the state is frozen. We keep a list of scheduled future events, always process the *earliest*, update statistics, and schedule what follows.

Two events only. **Arrival** (a customer appears) and **Departure** (the customer in service finishes).

State variables. clock t; server status $B \in \{0 \text{ idle}, 1 \text{ busy}\}$; number waiting Q (*excludes* the one in service); a FIFO list of the arrival times of those waiting; and t_{last} = time of the previous event.

Why / vs. the previous method

Next-event time advance vs. **fixed-increment**: a fixed-increment clock ticks by a constant Δt and asks "did anything happen?" every tick — most ticks are wasted, and two events inside one Δt get mis-ordered. Next-event jumps straight to the next event time, so it is both faster and exact. This is why all serious queueing simulation uses next-event.

The algorithm. Repeat until the stopping rule is met:

1. **Timing:** next event time = $\min(t_a, t_d)$ where t_a, t_d are the next scheduled arrival/departure. Advance t to it.
2. **Accumulate areas *before* changing state:**
$$\Delta t = t - t_{\text{last}}, \qquad t_{\text{last}} = t, \qquad A_Q \mathrel{+}= Q\,\Delta t, \qquad A_B \mathrel{+}= B\,\Delta t.$$
3. **Arrival:** schedule the next arrival. If server idle: delay = 0, record it, $B = 1$, schedule this customer's departure. If busy: $Q \mathrel{+}= 1$, push arrival time.
4. **Departure:** if $Q = 0$: $B = 0$, set $t_d = \infty$ (no delay recorded). Else: $Q \mathrel{-}= 1$, delay = t−(front arrival time), record it, schedule next departure.

Performance measures.
$$d(n) = \frac{1}{n}\sum_{i=1}^{n} D_i, \qquad q(n) = \frac{A_Q}{t} = \frac{1}{T}\int_0^T Q(\tau)\,d\tau, \qquad u(n) = \frac{A_B}{t} = \frac{1}{T}\int_0^T B(\tau)\,d\tau.$$

$d(n)$ is a plain *per-customer* mean of delays; $q(n)$ and $u(n)$ are *time-averages* — area under a step function divided by total time. (A queue of length 3 held for 2 hours ≠ length 3 held for 2 seconds, so we must weight by duration.)

Edge cases & traps

- **#1 killer:** update A_Q, A_B *before* mutating Q or B. The interval Δt just elapsed was spent at the *old* queue length / status.
- **Departure with empty queue:** set $B = 0$ and $t_d = \infty$; record *no* delay (else a phantom departure fires).
- **Tie $t_a = t_d$:** state a rule. The test "$t_a < t_d$" sends equality to the else-branch ⇒ *departure first* (frees the server so the arriver can enter service).
- **Balking / waiting-room limit k:** if $Q \geq k$ on arrival the customer leaves ("lost"); no delay, not counted in n.
- **First arrival** enters an idle server (delay 0); initialise $t_d = \infty$ so the first event is guaranteed an arrival.
- **Stopping rule:** "n delays completed" ≠ "n arrivals"; customers still waiting at stop time are not counted.

2 Linear Congruential Generator (LCG)

Recurrence.

$$X_{n+1} = (aX_n + c) \bmod m, \qquad U_n = \frac{X_n}{m} \in [0, 1).$$

a = multiplier, c = increment, m = modulus, X_0 = seed. **Mixed** LCG: $c \neq 0$. **Multiplicative** LCG (MCG): $c = 0$. The period is at most m (only m distinct states exist).

Why / vs. the previous method

The LCG replaces ad-hoc generators (like middle-square) because its behaviour is *fully predictable from number theory*: exact period, provable maximum-period conditions, and no absorbing "zero trap." It is strictly better than middle-square as a uniform source; its weakness is higher-dimensional structure (see RANDU).

Maximum-period conditions.

Case	Conditions	Max period
Mixed, $m = 2^b$, $c \neq 0$	$\gcd(c, m) = 1$ and $a \equiv 1 \pmod 4$	m
Multiplicative, $m = 2^b$, $c = 0$	seed odd and $a \equiv 3$ or $5 \pmod 8$	$m/4$
Multiplicative, m prime, $c = 0$	a is a primitive root mod m	$m - 1$

Hull–Dobell (mixed LCG reaches full period m) iff: (i) $\gcd(c, m) = 1$; (ii) $a - 1$ is divisible by every prime factor of m; (iii) $a - 1$ is divisible by 4 if $4 \mid m$.

The "good" generator (Lehmer / MINSTD): $a = 16807$, $c = 0$, $m = 2^{31} - 1$ (prime), period $= m - 1 \approx 2.1 \times 10^9$. Use this by default.

Edge cases & traps

- $c = 0$ with $X_0 = 0 \Rightarrow$ stuck at 0 (period 1); an MCG must never be seeded 0.
- $a = 1, c = 0 \Rightarrow$ constant (period 1). An even seed in the $m/4$ case falls into a shorter sub-cycle.
- **Passing a uniformity test $\neq$ a good generator. RANDU** ($a = 65539, c = 0, m = 2^{31}$) passes 1-D tests, yet every consecutive triple obeys $9X_i - 6X_{i+1} + X_{i+2} \equiv 0 \pmod{2^{31}}$ — all points lie on only 15 planes in 3-D.

3 Middle-Square Method & the Weyl Repair (MSWS)

Middle-square (d digits). Square the current value; zero-pad the square to $2d$ digits; take the middle d digits as the next value; divide by 10^d for U.

$$5731^2 = 32844361 \xrightarrow{\text{middle 4}} 8443 \longrightarrow 8443^2 = 71284249 \rightarrow 2842 \rightarrow \cdots$$

Why / vs. the previous method

Middle-square (von Neumann) is the *historical* generator kept only to show *why* we need better ones: squaring is not a bijection, so the sequence collapses into short cycles and can hit an absorbing zero. The LCG (§2) fixes this with a reversible modular map; MSWS (below) fixes it by bolting on a full-period Weyl term.

Edge cases & traps

- **Zero trap:** once the middle is $0 \ldots 0$ it stays there forever (absorbing state).
- **Short cycles:** many seeds collapse quickly (5731 falls into a 4-cycle).
- **Padding is mandatory:** without zero-padding to $2d$ digits you slice the wrong digits — a silent bug.
- **More digits does not fix it:** a d-digit state has only 10^d values and cycles appear near the birthday bound $\sqrt{10^d}$; bigger d only delays collapse.

Weyl repair (MSWS). Keep a second state w advancing by an odd constant s:

$$w_{n+1} = (w_n + s) \bmod 10^d, \qquad x_{n+1} = \left(\text{mid}(x_n^2) + w_{n+1}\right) \bmod 10^d.$$

The Weyl part has full period and never sticks at 0, so the pair (x, w) cannot collapse. The 32-bit exam version uses $s =$ `0xB5AD4EC5`, $m = 2^{32}$, output $U = \lfloor x/2^{16} \rfloor / 2^{16}$.

4 Testing Random Numbers

Two *independent* questions: (A) are the numbers **uniform**? — use χ^2 or K–S. (B) are they **independent**? — use autocorrelation or runs. A stream can pass one family and fail the other, so always test both.

4.1 Chi-Square (uniformity, frequency) Bin $[0,1)$ into k equal bins; expected count $E_i = N/k$; observed O_i.

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}, \qquad \text{df} = k-1, \qquad \text{reject } H_0 \text{ if } \chi_0^2 > \chi^2_{\text{crit}} \ (\text{df} = 9, \alpha = .05 : \ 16.919).$$

H_0: the numbers are Uniform$[0,1)$; a large χ_0^2 means the counts are far from flat.

Edge cases & traps

Need $E_i \geq 5$ (so $N \geq 5k$); bin index $\lfloor rk \rfloor$, clamp $r = 1.0$ down to $k-1$; for a *non-uniform* target use **equiprobable bins** from the inverse CDF (equal-width bins leave empty tail bins); and $p < \alpha \Rightarrow$ reject (do not flip it).

4.2 Kolmogorov–Smirnov (uniformity, CDF distance) Sort $R_{(1)} \leq \cdots \leq R_{(N)}$; compare the empirical CDF to the straight line of Uniform$[0,1]$:

$$D^+ = \max_{1 \leq i \leq N} \left(\tfrac{i}{N} - R_{(i)} \right), \quad D^- = \max_{1 \leq i \leq N} \left(R_{(i)} - \tfrac{i-1}{N} \right), \quad D = \max(D^+, D^-).$$

Large-N critical value ($\alpha = .05$): $D_{\text{crit}} = 1.36/\sqrt{N}$; reject H_0 if $D > D_{\text{crit}}$.

Why / vs. the previous method

K–S vs. χ^2: K–S works on the raw sorted sample (no binning), so it is better for **small** N and for exact CDFs; χ^2 needs many samples ($E_i \geq 5$) but handles **any** target distribution once you bin it. Both are *blind to order* — neither can see dependence.

Edge cases & traps

You must **sort first**; D^+ uses i/N but D^- uses $(i-1)/N$ (the classic slip); small N uses a table (e.g. $N = 5 \Rightarrow 0.565$). K–S is **blind to period**: a short-cycle MCG repeats the same values, whose marginal distribution still looks uniform, so K–S accepts it — catch it with a period check or an independence test.

4.3 Autocorrelation (independence) Tests whether values a lag ℓ apart are correlated ($H_0 : \rho = 0$). With start index i,

$$M = \left\lfloor \tfrac{N-i}{\ell} \right\rfloor - 1, \qquad \hat{\rho} = \frac{1}{M+1} \sum_{k=0}^{M} R_{i+k\ell} \, R_{i+(k+1)\ell} \ - \ 0.25,$$

$$\sigma_{\hat{\rho}} = \frac{\sqrt{13M+7}}{12(M+1)}, \qquad Z_0 = \frac{\hat{\rho}}{\sigma_{\hat{\rho}}}, \qquad \text{reject if } |Z_0| > 1.96 \ (\alpha = .05).$$

(M is the largest integer with $i + (M+1)\ell \leq N$.)

Edge cases & traps

Keep the -0.25 (it is $E[R_i R_j] = \frac{1}{4}$ for independent uniforms). **Test several lags**: a middle-square 4-cycle passes lag 1 and lag 5 but fails at **lag 4**. The formula's i is 1-based — offset by -1 in code.

4.4 Runs test (up/down, independence) Count a = number of monotone runs; under independence

$$\mu_a = \frac{2N-1}{3}, \qquad \sigma_a^2 = \frac{16N-29}{90}, \qquad Z_0 = \frac{a - \mu_a}{\sigma_a}, \qquad \text{reject if } |Z_0| > 1.96.$$

Why / vs. the previous method

Runs vs. autocorrelation: runs looks only at the *direction* of change (order pattern), so it catches trends and clustering cheaply without picking a lag; autocorrelation is sharper for *periodic* dependence but only at the lag you choose. Use runs as a quick order check, autocorrelation to hunt a suspected period.

Which test catches which defect (memorise):

Stream	χ^2 / K–S (uniform?)	Runs / Autocorr (independent?)
Good LCG (Lehmer)	Pass	Pass
Ramp $0, 0.001, \ldots$ (sorted)	**Pass** (perfectly flat)	**FAIL** (fully ordered)
Middle-square 4-cycle	Fail	Fail at lag 4 (passes 1 & 5)
$\sqrt{\text{uniform}}$	**FAIL** (density not flat)	Pass (order fine)

5 Monte Carlo Simulation

5.1 Sample-mean integration

$$I = \int_a^b g(x)\,dx \approx (b-a)\cdot\frac{1}{n}\sum_{i=1}^{n} g(X_i), \qquad X_i \sim \text{Uniform}(a,b).$$

Draw x uniformly, average the heights, multiply by the width.

5.2 Hit-or-miss estimation of π Drop (x, y) in the unit square; a hit is $x^2 + y^2 \le 1$ (inside the quarter circle of area $\pi/4$):

$$\hat{\pi} = 4\cdot\frac{\text{hits}}{n}.$$

Why / vs. the previous method

Sample-mean vs. hit-or-miss: both estimate the same area, but sample-mean averages a *continuous* height $g(X_i)$ while hit-or-miss averages a 0/1 indicator. The indicator has larger variance, so **sample-mean is usually more accurate for the same** n; hit-or-miss is used when the height is hard to evaluate but "inside/outside" is easy (e.g. π).

5.3 Buffon's needle Parallel lines a distance D apart, needle length $L \le D$:

$$P(\text{cross}) = \frac{2L}{\pi D} \;\Rightarrow\; \hat{\pi} = \frac{2L}{D\,\hat{P}}, \qquad x \sim U\big(0, \tfrac{D}{2}\big)\,,\ \theta \sim U\big(0, \tfrac{\pi}{2}\big)\,,\ \text{cross} \iff x \le \tfrac{L}{2}\sin\theta.$$

Edge cases & traps

Use **two** independent uniforms per needle (x and θ). If $\hat{P} = 0$ (tiny n), $\hat{\pi}$ blows up. A *short RNG period* means more samples just recycle the same numbers — the error plateaus and cannot improve.

5.4 Error & convergence

$$\text{SE} \propto \frac{1}{\sqrt{n}} \quad (\text{halve the error} \Rightarrow 4\times \text{ the samples}), \qquad 95\%\ \text{CI} = \hat{I} \pm 1.96\,\text{SE}, \qquad \text{SE}_{\text{hit-miss}} = 4\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Why / vs. the previous method

Monte Carlo vs. Simpson's rule: Simpson is $O(n^{-4})$ in 1-D — far better — but needs n^d grid points in d dimensions and becomes hopeless past $d \approx 4$. Monte Carlo's $O(n^{-1/2})$ is *independent of dimension*, so it wins for high-dimensional integrals. Slow but dimension-proof.

5.5 Reliability networks Component lifetimes T_i are drawn by inverse-transform, then combined by topology:

$$\text{series (both needed): } T = \min(T_a, T_b), \qquad \text{parallel (either works): } T = \max(T_a, T_b),$$

$$\hat{p} = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}[\,Y_k \ge \theta\,] \quad (\text{fraction surviving past time } \theta).$$

Edge cases & traps

series = min, **parallel** = max (do not swap). Reduce the topology inside-out. A spare helps most at the current **bottleneck**. Use a fixed stream assignment (e.g. $U_i = U_{4k+i}$) so replications stay independent and reproducible.

6 Inverse-Transform Variate Generation

Draw $U \sim \text{Uniform}(0,1)$ and return $X = F^{-1}(U)$, i.e. solve $U = F(x)$ for x.

Distribution	**CDF** $F(x)$	**Inverse** $X = F^{-1}(U)$
Uniform(a,b)	$(x-a)/(b-a)$	$a + (b-a)U$
Exponential (mean β)	$1 - e^{-x/\beta}$	$-\beta \ln(1-U)$
Weibull(k,c)	$1 - e^{-(x/c)^k}$	$c\big(-\ln(1-U)\big)^{1/k}$
Triangular(a,b,c)	piecewise	$U \le \frac{c-a}{b-a}$: $a + \sqrt{U(b-a)(c-a)}$; else $b - \sqrt{(1-U)(b-a)(b-c)}$
Discrete	staircase	smallest value whose cumulative prob $\ge U$

Why / vs. the previous method

Inverse-transform is the default because it is *exact* and uses **one** uniform per variate (efficient, easy to synchronise across replications). It needs a closed-form F^{-1}; when none exists (e.g. Normal) one switches to acceptance–rejection or a special algorithm. It only *maps* randomness — it never creates any.

Edge cases & traps

Use $1-U$ inside the log and never let $U = 0$ ($\ln 0 = -\infty$); since $U \in [0,1)$, $1-U \in (0,1]$ is safe. A short-period or correlated input stream makes the variates inherit the same flaws. Verify by **moments**: Exp(β) has mean β, variance β^2; Uniform(a,b) has variance $(b-a)^2/12$.

7 Solved Simulation Problems (math practice)

P1 — LCG period & Hull–Dobell. For $X_{n+1} = (5X_n + 3) \bmod 16$, $X_0 = 7$, find the period. Then decide whether $a = 7, c = 3$ gives full period.

Solution. Iterating: $7, 6, 1, 8, 11, 10, 5, 12, 15, 14, 9, 0, 3, 2, 13, 4$, and $X_{16} = (5 \cdot 4 + 3) \bmod 16 = 23 \bmod 16 = 7 = X_0$. All 16 residues appear, so **period** $= 16$ **(full)**. Check Hull–Dobell for $a = 5$: $\gcd(3, 16) = 1$✓; $m = 16 = 2^4$ has only prime factor 2, and $a - 1 = 4$ is divisible by 2✓; $4 \mid m$ so need $4 \mid (a-1)$: $4 \mid 4$✓⇒ full period, confirmed. For $a = 7$: $a - 1 = 6$ is *not* divisible by 4, so condition (iii) fails ⇒ **not** full period.

P2 — Chi-square test. $N = 100$ numbers fall into $k = 10$ bins with observed counts $O = [8, 10, 9, 12, 11, 7, 13, 10, 9, 11]$. Test uniformity at $\alpha = 0.05$.

Solution. $E_i = N/k = 10$ (and $E_i \ge 5$✓). Deviations^2: $4, 0, 1, 4, 1, 9, 9, 0, 1, 1$, sum $= 30$.

$$\chi^2_0 = \frac{30}{10} = 3.0, \qquad \text{df} = k - 1 = 9, \qquad \chi^2_{\text{crit}} = 16.919.$$

$3.0 < 16.919 \Rightarrow$ **do not reject** H_0; the numbers are consistent with Uniform$[0,1)$.

P3 — Kolmogorov–Smirnov test. $R = \{0.44, 0.81, 0.14, 0.93, 0.05\}$, $N = 5$, $\alpha = 0.05$.

Solution. Sort: $0.05, 0.14, 0.44, 0.81, 0.93$.

i	1	2	3	4	5
$i/N - R_{(i)}$	0.15	0.26	0.16	−0.01	0.07
$R_{(i)} - (i-1)/N$	0.05	−0.06	0.04	0.21	0.13

$D^+ = 0.26$, $D^- = 0.21$, $D = \max = 0.26$. Table value $D_{\text{crit}}(N = 5) = 0.565$. Since $0.26 < 0.565 \Rightarrow$ **do not reject**; uniform.

P4 — Autocorrelation. $R = \{0.2, 0.5, 0.9, 0.3, 0.6, 0.1\}$ ($N = 6$), test lag $\ell = 1$ from start $i = 1$.

Solution. Largest M with $1 + (M+1) \cdot 1 \le 6$ gives $M = 4$. Products $R_k R_{k+1}$: 0.10, 0.45, 0.27, 0.18, 0.06; sum = 1.06.

$$\hat{\rho} = \frac{1.06}{M+1} - 0.25 = \frac{1.06}{5} - 0.25 = 0.212 - 0.25 = -0.038, \quad \sigma = \frac{\sqrt{13 \cdot 4 + 7}}{12 \cdot 5} = \frac{\sqrt{59}}{60} = 0.128.$$

$Z_0 = -0.038/0.128 = -0.30$. $|Z_0| = 0.30 < 1.96 \Rightarrow$ **do not reject**; no lag-1 dependence detected.

P5 — Monte Carlo integration. Estimate $\int_0^1 x^2\,dx$ (true = 1/3) with the five uniforms $U = \{0.1, 0.3, 0.5, 0.7, 0.9\}$.

Solution. Here $a = 0, b = 1$, $g(x) = x^2$. Heights: $0.01, 0.09, 0.25, 0.49, 0.81$; mean $= 1.65/5 = 0.33$.

$$\hat{I} = (b-a) \cdot \overline{g} = 1 \cdot 0.33 = 0.33, \qquad \text{error} = |0.3333 - 0.33| \approx 0.003.$$

P6 — Hit-or-miss π with confidence interval. Of $n = 10$ points in the unit square, 8 land inside the quarter circle.

Solution. $\hat{p} = 0.8$, $\hat{\pi} = 4\hat{p} = 3.2$.

$$\text{SE} = 4\sqrt{\frac{0.8 \cdot 0.2}{10}} = 4\sqrt{0.016} = 4(0.1265) = 0.506, \quad 95\%\ \text{CI} = 3.2 \pm 1.96(0.506) = 3.2 \pm 0.99.$$

So $\hat{\pi} \in [2.21, 4.19]$ — wide, because n is tiny (the $1/\sqrt{n}$ law).

P7 — Buffon's needle. $L = 1$, $D = 2$, $N = 1000$ needles, 310 crossings. Estimate π.

Solution. $\hat{P} = 310/1000 = 0.31$.

$$\hat{\pi} = \frac{2L}{D\,\hat{P}} = \frac{2(1)}{2(0.31)} = \frac{1}{0.31} = 3.226.$$

(True crossing prob is $2L/(\pi D) = 1/\pi \approx 0.318$, consistent with $\hat{P} = 0.31$.)

P8 — Inverse-transform variates. With $\beta = 2$ and $U = 0.25$ generate an Exponential; with $k = 2, c = 3, U = 0.5$ generate a Weibull.

Solution.

$$X_{\text{exp}} = -\beta \ln(1-U) = -2\ln(0.75) = -2(-0.2877) = 0.575,$$

$$X_{\text{wei}} = c(-\ln(1-U))^{1/k} = 3(-\ln 0.5)^{1/2} = 3(0.6931)^{0.5} = 3(0.8326) = 2.498.$$

P9 — Discrete-event hand simulation. Single server, FIFO. Four customers arrive at $t = 0, 1, 2, 6$ with service times $S = [3, 2, 1, 2]$. Tie rule: departure before arrival. Find $d(4), q(4), u(4)$.

Solution. Event trace (B=status, Q=waiting):

t	Event	B	Q	Note
0	C1 arrives	1	0	delay 0; depart at 3
1	C2 arrives	1	1	waits
2	C3 arrives	1	2	waits
3	C1 departs	1	1	C2 starts, delay $3-1=2$; depart at 5
5	C2 departs	1	0	C3 starts, delay $5-2=3$; depart at 6
6	C3 departs, then C4 arrives	1	0	tie: depart first; C4 delay 0; depart at 8
8	C4 departs	0	0	stop, $T = 8$

Delays $= [0, 2, 3, 0] \Rightarrow d(4) = 5/4 =$ **1.25**. Areas: $A_Q = \int Q\,dt = (0)(1) + (1)(1) + (2)(1) + (1)(2) + (0)(1) +$

$(0)(2) = 5$; the server is busy the whole $[0, 8]$ so $A_B = 8$.

$$q(4) = \frac{A_Q}{T} = \frac{5}{8} = \mathbf{0.625}, \qquad u(4) = \frac{A_B}{T} = \frac{8}{8} = \mathbf{1.0}.$$

P10 — Reliability of a mixed network. System = (A series B) parallel C; each lifetime is Exponential with mean 100. Using $U_A = 0.5, U_B = 0.2, U_C = 0.9$, does the system survive to $\theta = 50$?

Solution. Inverse-transform lifetimes ($T = -100 \ln(1 - U)$):

$$T_A = -100 \ln 0.5 = 69.3, \quad T_B = -100 \ln 0.8 = 22.3, \quad T_C = -100 \ln 0.1 = 230.3.$$

Series block: $T_{AB} = \min(69.3, 22.3) = 22.3$. Whole system: $T = \max(T_{AB}, T_C) = \max(22.3, 230.3) = 230.3$. Since $230.3 \geq \theta = 50$, the system **survives** this replication ($\mathbf{1}[T \geq 50] = 1$). Over N replications, $\hat{p} = \frac{1}{N} \sum \mathbf{1}[T_k \geq 50]$. Note B is the bottleneck (smallest T), so a spare in *parallel with* B would help the series path most.